

CSE342

Statistical Machine Learning

Assignment-3

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Question: 1

(60000, 28, 28)

(60000,)

(10000, 28, 28)

(10000,)

(18623, 784)

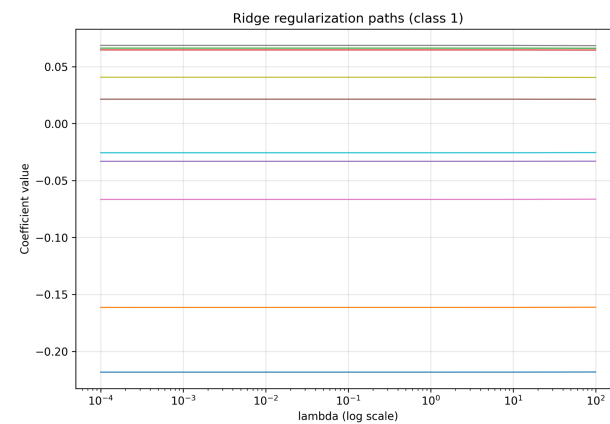
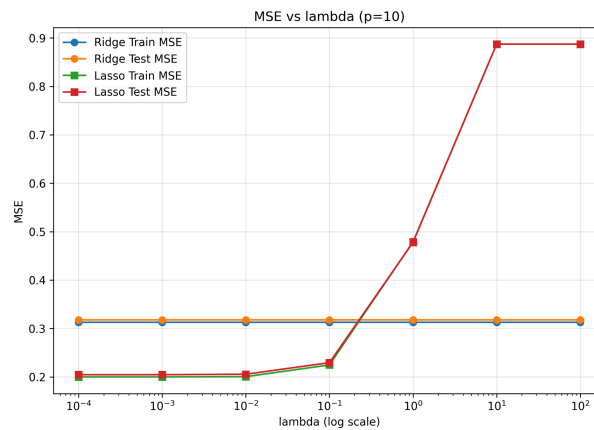
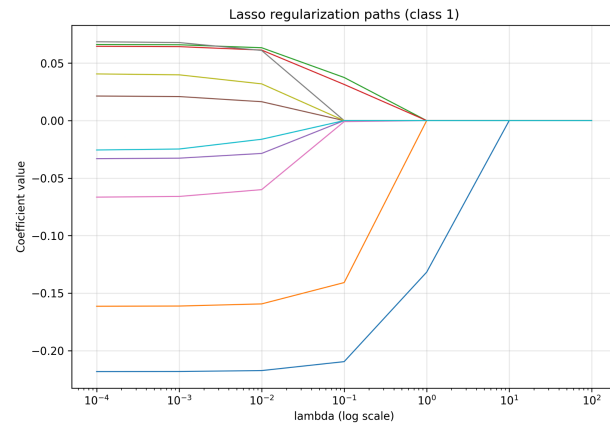
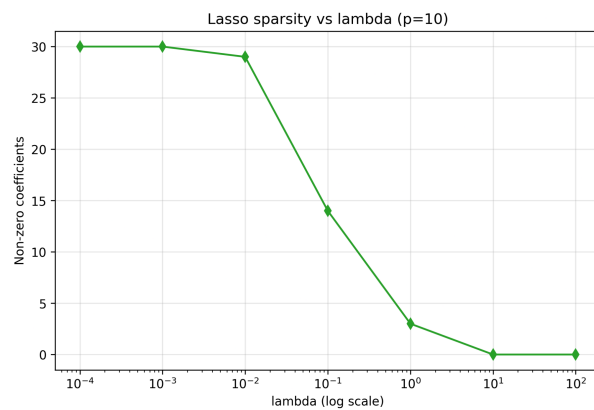
(18623,)

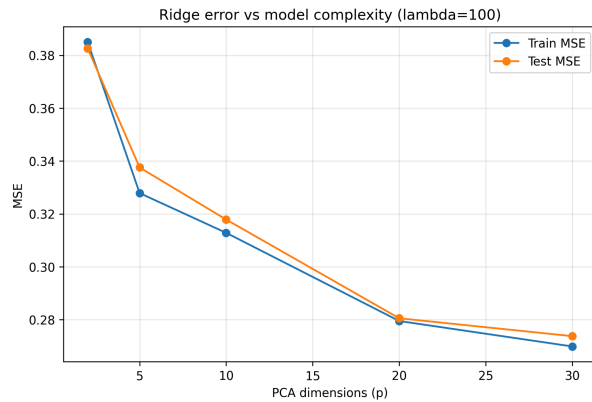
Best Ridge lambda (p=10): 100

Best Lasso lambda (p=10): 0.0001

Best Ridge test accuracy: 95.14%

Best Lasso test accuracy: 94.76%





Observation:

Ridge regression MSE was quite stable, with no significant changes in any of the lambda values that were tested, and was close to 0.30 -0.31 on both training and test sets. This stability is typical of ridge: L2 regularisation smoothly shrinks all coefficients to zero without assigning them to zero, meaning that the model will still have predictive power at large lambda. At lambda = 100 (Best Ridge test accuracy: 95.14 percent), the test MSE is minimized, which may be counterintuitive but tends to indicate that strong regularisation of L2 in a 10-dimensional PCA-reduced space is preventing overfitting.

Lasso train and test MSE were close to ridge levels (lambda = 0.01) up to lambda = 0.1, but thereafter test MSE shot to high values (around 0.9 at lambda = 100). This is consistent with the familiar behavior of Lasso: at smaller lambdas, it performs just as well as ridge, but at larger lambdas, it aggressively zeros out coefficients, ultimately underfitting. The best Lasso lambda was 0.0001 with a test accuracy of 94.76.

Question: 2

Data after filtering classes {0,1,2}, flattening, normalization, and PCA (p=10):

Train shape: (18623, 10), Test shape: (3147, 10)

Custom Decision Tree (3 terminal nodes)

Overall accuracy: 78.17%

Class 0 accuracy: 77.55%

Class 1 accuracy: 99.74%

Class 2 accuracy: 55.04%

Bagging average OOB error (5 trees): 0.2110

Bagging Test Performance

Overall accuracy: 78.14%

Class 0 accuracy: 77.35%

Class 1 accuracy: 99.65%

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Class 2 accuracy: 55.23%
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k=1 , Avg OOB Error: 0.4717
k=2 , Avg OOB Error: 0.3594
k=3 , Avg OOB Error: 0.3988
k=4 , Avg OOB Error: 0.3533
k=5 , Avg OOB Error: 0.3556
k=6 , Avg OOB Error: 0.2984
k=7 , Avg OOB Error: 0.2555
k=8 , Avg OOB Error: 0.2158
k=9 , Avg OOB Error: 0.2143
k=10 , Avg OOB Error: 0.2110

Optimal k : 10 , OOB Error: 0.2110)
Random Forest (Best k=10) Test Performance
Overall accuracy: 78.14%
Class 0 accuracy: 77.35%
Class 1 accuracy: 99.65%
Class 2 accuracy: 55.23%
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Observation

Threshold justification: The median of each feature's values was used as the candidate split threshold. The median is preferable to the mean because it is robust to long tails that may occur in PCA-projected distributions, and it ensures a non-trivial split (the two child nodes receive at least a significant fraction of samples) even with a skewed distribution.

In the initial split, the weighted Gini index of all 10 PCA dimensions was considered. The dimension with the least weighted Gini was chosen. The resulting tree was generated with 3 terminal nodes (one more split on the better child) as needed. The results of the accuracy are depicted.

Optimal $k = 10$ (i.e., using all the features) minimized OOB error. This implies that when using this tree depth and input feature set, restricting the feature subset is not beneficial; it only increases bias without significantly reducing inter-tree correlation. Probably due to the limited number of PCA features (10) and depth-2 trees, there is already minimal redundancy to be used by random subspace selection.

Random Forest is not working optimally here; it is equal to bagging at $k = 10$. Random Forest works best when trees are deep (high variance) and when there are many/correlated features. In this case, shallow trees and a small feature set imply that feature subsampling causes noise rather than diversity.

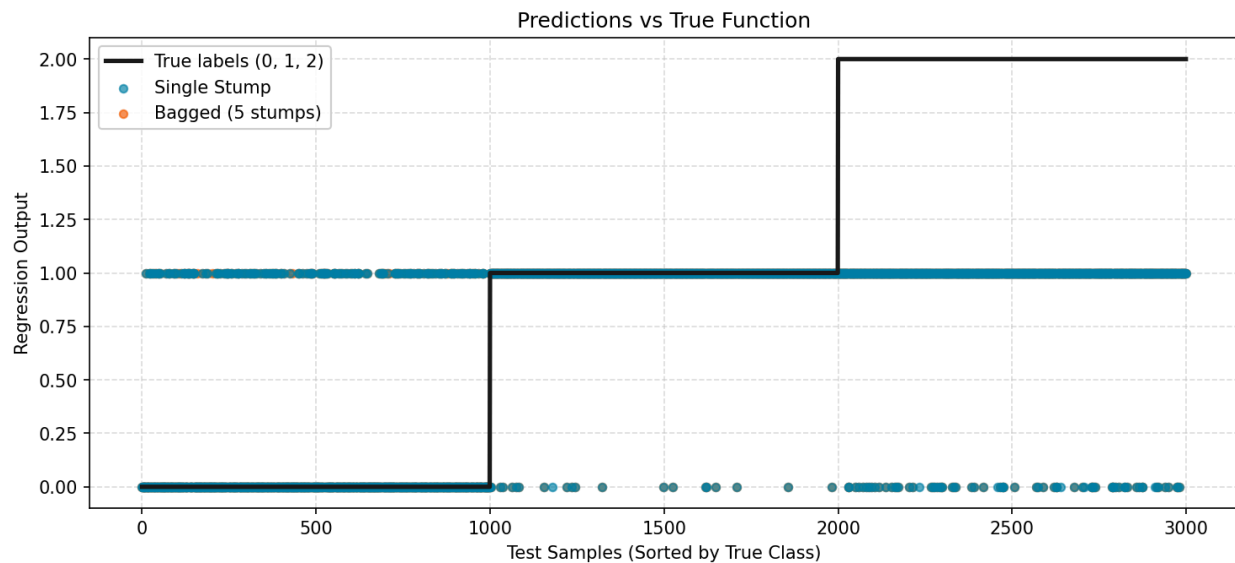
Question: 3

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Train shape after filtering: (18000, 10)
Test shape after filtering: (3000, 10)
Train/Test shape after PCA: (18000, 10) (3000, 10)
Top 10 eigenvalues:
[18.95839226  8.2487155  5.38427381  2.16459503  1.38624661  1.35265381
 1.08617691  0.85694578  0.81091662  0.67330796]
```

```
Single regression decision stump
Best feature index: 2
Best threshold: -1.450796587671947
Left region prediction: 0.15961606825453253
Right region prediction: 1.382091482139971
Minimum SSR on train set: 6220.136243149647
Test MSE: 0.3629439763726561
Model 1 OOB MSE: 0.3432667576471788
Model 2 OOB MSE: 0.346762192975777
Model 3 OOB MSE: 0.35028460756834656
Model 4 OOB MSE: 0.36006633928209847
Model 5 OOB MSE: 0.3423072514434691
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Bagging with 5 regression decision stumps
Average OOB MSE: 0.348537429783374
Bagging test MSE: 0.35737421310293804
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Q3 - Regression Stump vs Bagging on Fashion-MNIST



Observation

The optimal split point on feature 2 was identified by a regression decision stump with the best MSE of 0.3629 at a threshold of 10 PCA features, corresponding to a value of 1.451. A stump only produces two output values (~ 0.16 and ~ 1.38), and it is therefore unable to differentiate between all three different classes, with all samples of class 2

being nearly completely misclassified as class 1. Gaining more than 5 bootstrap stumps dropped the test MSE to 0.3574 and the average OOB MSE of 0.3485, which was in close correspondence with generalization. The enhancement is, however, slight as the bottleneck is high bias, not variance - averaging stumps that all have the same structural constraint bring little variation. It would require a more elaborate tree to meaningfully divide all three classes.